Design Documentation:

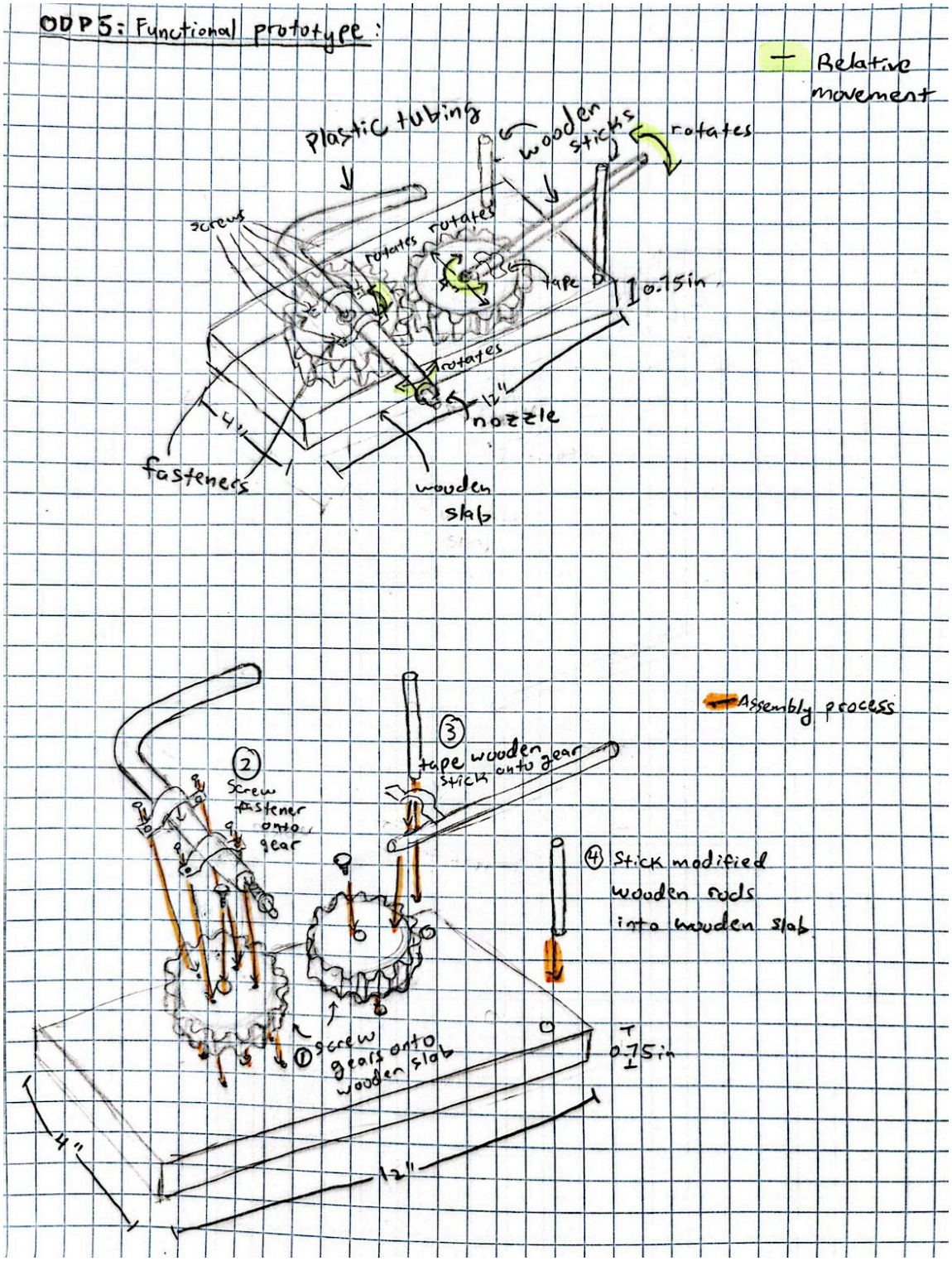
(Parts List) MAE2250 Parts List

	A	B	C	D	E	F	G	H	I
1	**Part Name**	**McMaster Code**	**Quantity**	**Unit Cost**	**Total Cost**	**Link(s)**	**Notes**	**Ordered**	**Delivered**
2	Steel Hex Head Screw for Wood Zinc-Plated, M12 Screw Size, 45 mm Long	97498A172	1	$9.43	$9.43	https://www.mcmaster.com/97	Pack of 10	☑	☐
3	Vibration-Damping Routing Clamp Zinc-Plated Steel with TPE Rubber Cushion, 7/8" ID	11355T26	2	$4.15	$8.30	https://www.mcmaster.com/11	1 part	☑	☑
4	Black-Oxide Alloy Steel Socket Head Screw 10-24 Thread Size, 5/16" Long	91864A032	1	$6.12	$6.12	https://www.mcmaster.com/91	Pack of 10	☑	☑
5	4" diameter PLA printed Gears		2	$3.99	$7.98	https://www.mcmaster.com/60	**RPL Printed**	☑	☑
6	Soft Masterkleer PVC Tubing for Air and Water, 1" ID, 1-1/2" OD	5233K48	2	$9.88	$19.76	https://www.mcmaster.com/52	**Found in Recyl**	☑	☑
7	Push-to-Connect Fitting for Drinking Water, Adapter, 1/4" Tube Stem OD, for 3/16" Barbe	51055K57	1	$2.21	$2.21	https://www.mcmaster.com/51	**Found in Recyl**	☑	☑
8	12" * 12" Wooden Slabs		1	$3.00	$3.00	N/A	**Taylor design s**	☑	☑
9	24" Wooden Shaft length, 1/4 inch diameter		1	$3.00	$3.00	N/A	**Taylor design s**	☑	☑
10	Tape		1	$0.00	$0.00	N/A	**Taylor design s**	☑	☑
11	**Total**				$59.80			☑	☑

Fabrication of parts list:

1. Cut the 12-inch by 12-inch wooden slab into three equal-width sections
2. Glued them together with wood glue and clamped them together at the four corners allowed it to rest for 24 hours.
3. Cut the tube down to roughly an 8-inch length, just to make sure there was not too much excess, making the mechanism difficult to use.
4. Cut down the 24-inch length, wooden shaft into four 6-inch pieces to prevent motion.
5. Cut one more wooden shaft to roughly 8 inches, and attach it to one gear using tape to use it as a handle.
6. 3D printed gears with CAD from McMaster at the RPL, slightly edited CAD to add a larger hole in the center to decrease the rotational inertia of the mechanism.
7. Drill a 27/64 inch drill bit into the wooden base and glue it down to ensure it is secure. Push two wooden shafts through the two 3D printed gears, but allow the gears to freely rotate. (This was supposed to be down with wooden screws, but our part didn't come)
8. Attach the modified 8-inch tubing horizontally to one gear with tape. (This was supposed to be drilled down, but out clamp were a different dimension than intended)
9. Attach the 8-inch wooden shaft horizontally to the other gear with tape. This will be used for control of the rotational motion of the gears.
10. Drilled a ¼ inch drill bit on each side of the 8-inch wooden shaft. Glue down the two 6-inch wooden shafts into the two holes.

(Illustration)



Assembly Instructions:

We need arrows in assembly for movement

1) Fasten gears to wooden board with screws so gears so gears are toothed toigether

2) Fasten tube for gas dispersion to one gear
3) Fasten handle to the other gear to allow for controlled rotation
4) Attach wooden shafts into wooden board to constrain the rotation of the gears

Design vs. Drawing Discrepancies:

1) One major discrepancy is our use of tape as a fastener. This is due to the fact that the clamps we ordered were much larger than we expected, and ended up not being useful for us. This mistake was due to a miscommunication of the tube's radius and diameter. The team has worked this out, and we have decided to always communicate using diameter due to that being the engineering standard.
2) Another discrepancy is our lack of wood screws through the center of our gears. This is due to our screw order getting delayed. Although unfortunate, we can prevent this in the future by pushing forward our ordering deadlines on our schedule.

Design Test:

Test 1: Rotation Smoothness
Part tested: Gears
What it tests for: Whether the handle-driven gears rotate the nozzle smoothly without binding, slipping, or excessive friction
How we tested it: Handle was rotated through desired range of motion while observing gear motion and checking for slip, jerky motion, or interference
Test results: Before applying constraints to the range of motion, we rotated the gears through 20 full rotations of 360 degrees. Throughout the test, there was no slippage or binding of the gear teeth.
Conclusion for next iteration: While we did not see any slipping/binding, we noted that the gears must be perfectly placed to achieve that. To provide slight room for error, if we re-fabricate the gears we will increase the gear tooth height by 5mm.

Test 2: Aiming Range
Part tested: Tube/Nozzle
What it tests for: Whether gear system allows tube/nozzle to rotate through motion required to cover desired volumetric space by the gas
How we tested it: Handle was turned from one extreme to the other and total angular rotation of the nozzle was measured
Test results: The gear system completed 10 full back-and-forth cycles. With the constraints we applied, we achieved a range of motion of 90 degrees. There were 0 instances of slipping or binding by the gear teeth during the test, and the nozzle was able to achieve the range of motion without exhibiting any significant stress.
Conclusion for next iteration: Our method of constraining the gears was effective. However, to make the design more robust, we will increase the diameter of the wooden cylinder constraints and use an aluminum base instead of wood.

Test 3: Repeated-Use Durability
Part tested: Screws/Gears/Wooden Board assembly

What it tests for: If mechanism maintains functionality after repeated use without loosening, slipping, or increasing friction
How we tested it: Handle was cycled at a minimum of 20 times while observing changes in motion smoothness, gear interface, and structural stability
Test results: We cycled the gear through 20 full cycles at 20 RPM once the design was fully constructed, including constraints. The design held up with no clear signs of wear and tear.
Conclusion for next iteration: While out initial design holds up, it is primarily constructed out of cheap wood and tape, meaning that it will lose significant strength over time. For our final design, we will convert the constraint shafts and the base to aluminum. We will also use screws to keep the gears in place instead of a wooden sticks.

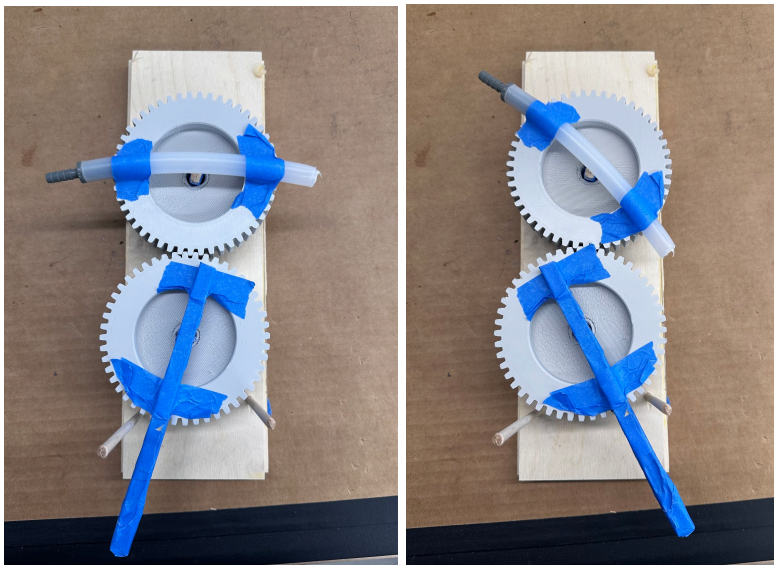
Test 4: Tube Stability
Part tested: Pipe Clamp
What it tests for: Whether gas tube remains securely attached and doesn't interfere with nozzle rotation during operation
How we tested it: Rotate tube through full range of motion while observing clamps, checking for twisting, sagging, or resistance to rotation.
Test results: Unfortunately we were not able to directly test this due to our clamp ordering mishap. We however can confidently say that the clamps would hold up in the design, because we taped evetrything down with painter's tape, which is significantly weaker. Through all of our previous rotations, roughly 20 full cycles at varying speeds, the tape held up just fine. This makes us feel confident to say our clamps would hold up fine, due to the strength of all portions of the clamps being orders of magnitude stronger than the tape. The tape also did not prevent rotation, which means the clamps would not as well due to it being in a similar position and not interfering with the gear teeth.
Conclusion for next iteration: We are confident that once we have the correct clamps, they will be strong enough to hold our tube and handle down to demonstrate functionality of our design.

Test 5: Structural Stability
Part tested: Wood Mounting Board
What it tests for: If wooden base remains rigid and maintains gear alignment during use
How we tested it: Handle was rotated repeatedly while checking for board flexing, shifting, or misalignment of the gears
Test results: We rotated the board through 10 full cycles at 20 RPM while the tube was held down to simulate stress from the harvester, applying a force on the mechanism of 10 Newtons. The wooden base exhibited no signs of flexing/bending, and the gears remained tightly connected with no slippage.
Conclusion for next iteration: The fact that the base did not flex or shift shows that our design does not have misalignment or place significant stress on the base. That being said, we still plan to use an aluminum base in the next iteration to ensure durability over more extended periods of time.

Success Criteria:

Our main goal of this iteration is to ensure our gear rotation system for gas dispersion is effective and is durable enough to remain undamaged. A successful test should follow all of our outlined design criteria, and effectively demonstrate the ability to move our nozzle in a periodic type of motion to achieve better gas dispersion.

- After repeated and continuous testing for over 1 minute, where gears spin at least 90 degrees, all gears still spin as intended without interference or skipping
    - The gears would need to spin well for the entirety of the harvesting process for multiple harvests. This is important because we would want our solution to work long term for the farmers, so they don't have to do continuous maintenance of our lanternfly gas dispersal system. An effective lanternfly prevention system is durable, so for our solution to work well we would need it to not only effectively deter lanternflies away from grapes, but also be durable enough to last a long time.
- After repeated and continuous testing for over 1 minute, where gears spin at least 90 degrees, all parts remain structurally stable/intact without breaking apart
    - The gears would need to spin well for the entirety of the harvesting process for multiple harvests. This is important because we would want our solution to work long term for the farmers, so they don't have to do continuous maintenance of our lanternfly gas dispersal system. An effective lanternfly prevention system is durable, so for our solution to work well we would need it to not only effectively deter lanternflies away from grapes, but also be durable enough to last a long time.
- Gears that are constrained such that the gas dispersion tube only has an approximate 90-degree range of motion without experiencing significant resistance forces (friction, etc.)
    - Effective gas dispersion is crucial for our project to work, because it theoretically means the lanternflies would be better dispersed away from the grapes due to a lack of oxygen.

<u>Demo</u>

Upon spinning the handle, the nozzle rotates with a 90 degrees free rotation due to the smooth rotation of the spur gears connected underneath.